

# HW5

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2026-06-07

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.0      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.3      v tibble    3.3.1
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr      1.2.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
library(skimr)
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
# a. Load Data
df <- read.csv("sp500_companies.csv")

# drop long text column
df <- df %>% select(-Longbusinesssummary)
dim(df)
```

```
## [1] 502  15
```

```
str(df)
```

```
## 'data.frame':    502 obs. of  15 variables:
## $ Exchange      : chr  "NMS" "NMS" "NMS" "NMS" ...
## $ Symbol        : chr  "AAPL" "NVDA" "MSFT" "AMZN" ...
## $ Shortname     : chr  "Apple Inc." "NVIDIA Corporation" "Microsoft Corporation" "Amazon.com, Inc." ...
## $ Longname      : chr  "Apple Inc." "NVIDIA Corporation" "Microsoft Corporation" "Amazon.com, Inc." ...
## $ Sector        : chr  "Technology" "Technology" "Technology" "Consumer Cyclical" ...
## $ Industry      : chr  "Consumer Electronics" "Semiconductors" "Software - Infrastructure" "Internet & Telecom" ...
## $ Currentprice   : num  254 135 437 225 191 ...
## $ Marketcap     : num  3.85e+12 3.30e+12 3.25e+12 2.37e+12 2.35e+12 ...
```

```
## $ Ebitda : num 1.35e+11 6.12e+10 1.37e+11 1.12e+11 1.23e+11 ...
## $ Revenuegrowth : num 0.061 1.224 0.16 0.11 0.151 ...
## $ City : chr "Cupertino" "Santa Clara" "Redmond" "Seattle" ...
## $ State : chr "CA" "CA" "WA" "WA" ...
## $ Country : chr "United States" "United States" "United States" "United States" ...
## $ Fulltimeemployees: int 164000 29600 228000 1551000 181269 181269 72404 140473 20000 396500 ...
## $ Weight : num 0.0692 0.0593 0.0584 0.0425 0.0423 ...
```

```
head(df)
```

```
## Exchange Symbol Shortname Longname
## 1 NMS AAPL Apple Inc. Apple Inc.
## 2 NMS NVDA NVIDIA Corporation NVIDIA Corporation
## 3 NMS MSFT Microsoft Corporation Microsoft Corporation
## 4 NMS AMZN Amazon.com, Inc. Amazon.com, Inc.
## 5 NMS GOOGL Alphabet Inc. Alphabet Inc.
## 6 NMS GOOG Alphabet Inc. Alphabet Inc.
## Sector Industry Currentprice
## 1 Technology Consumer Electronics 254.49
## 2 Technology Semiconductors 134.70
## 3 Technology Software - Infrastructure 436.60
## 4 Consumer Cyclical Internet Retail 224.92
## 5 Communication Services Internet Content & Information 191.41
## 6 Communication Services Internet Content & Information 192.96
## Marketcap Ebitda Revenuegrowth City State Country
## 1 3.846820e+12 1.34661e+11 0.061 Cupertino CA United States
## 2 3.298803e+12 6.11840e+10 1.224 Santa Clara CA United States
## 3 3.246069e+12 1.36552e+11 0.160 Redmond WA United States
## 4 2.365034e+12 1.11583e+11 0.110 Seattle WA United States
## 5 2.351625e+12 1.23470e+11 0.151 Mountain View CA United States
## 6 2.351623e+12 1.23470e+11 0.151 Mountain View CA United States
## Fulltimeemployees Weight
## 1 164000 0.06920915
## 2 29600 0.05934964
## 3 228000 0.05840088
## 4 1551000 0.04254995
## 5 181269 0.04230871
## 6 181269 0.04230867
```

```
# b. Exploration
```

```
skim(df)
```

Table 1: Data summary

|                        |     |
|------------------------|-----|
| Name                   | df  |
| Number of rows         | 502 |
| Number of columns      | 15  |
| Column type frequency: |     |
| character              | 9   |
| numeric                | 6   |

|                 |      |
|-----------------|------|
| Group variables | None |
|-----------------|------|

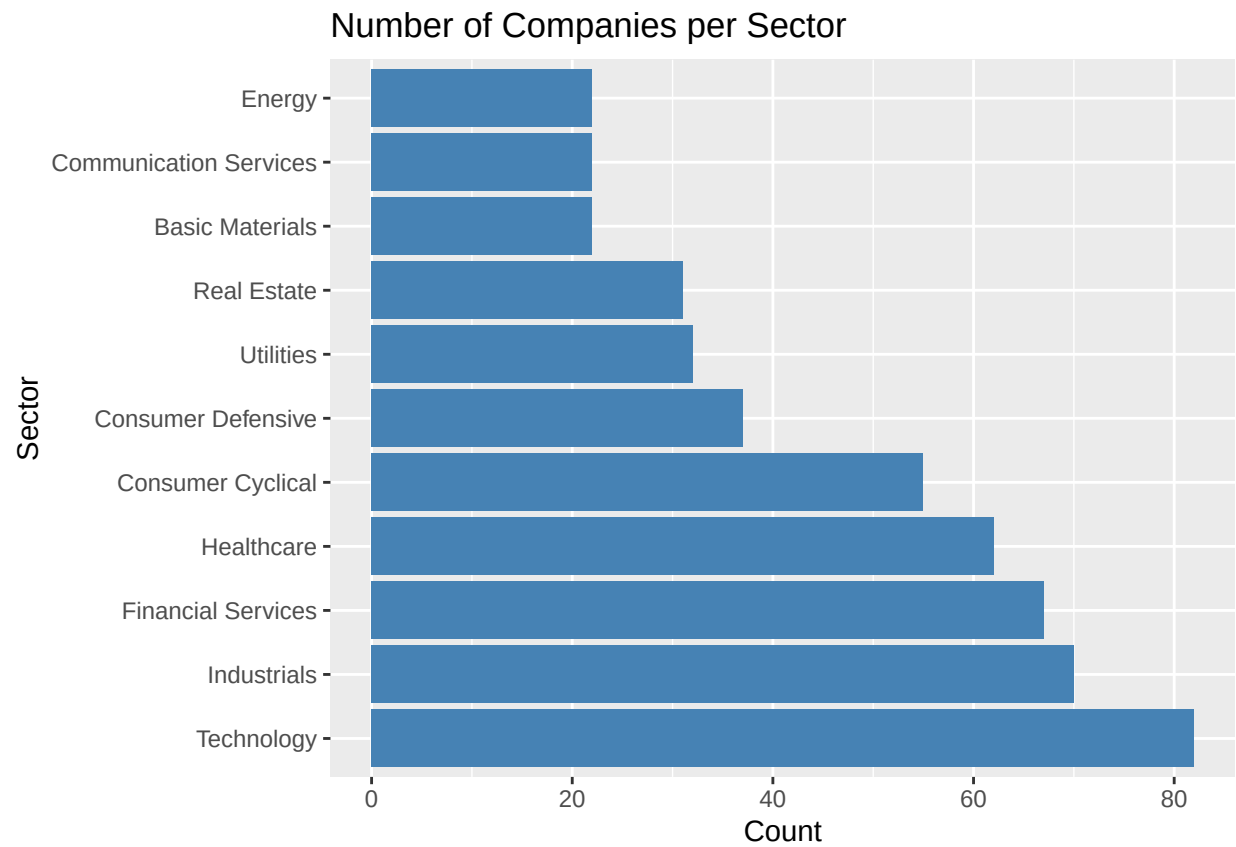
### Variable type: character

| skim_variable | n_missing | complete_rate | min | max | empty | n_unique | whitespace |
|---------------|-----------|---------------|-----|-----|-------|----------|------------|
| Exchange      | 0         | 1             | 3   | 3   | 0     | 4        | 0          |
| Symbol        | 0         | 1             | 1   | 5   | 0     | 502      | 0          |
| Shortname     | 0         | 1             | 7   | 31  | 0     | 499      | 0          |
| Longname      | 0         | 1             | 6   | 47  | 0     | 499      | 0          |
| Sector        | 0         | 1             | 6   | 22  | 0     | 11       | 0          |
| Industry      | 0         | 1             | 4   | 40  | 0     | 114      | 0          |
| City          | 0         | 1             | 4   | 18  | 0     | 235      | 0          |
| State         | 0         | 1             | 0   | 2   | 20    | 42       | 0          |
| Country       | 0         | 1             | 6   | 14  | 0     | 8        | 0          |

### Variable type: numeric

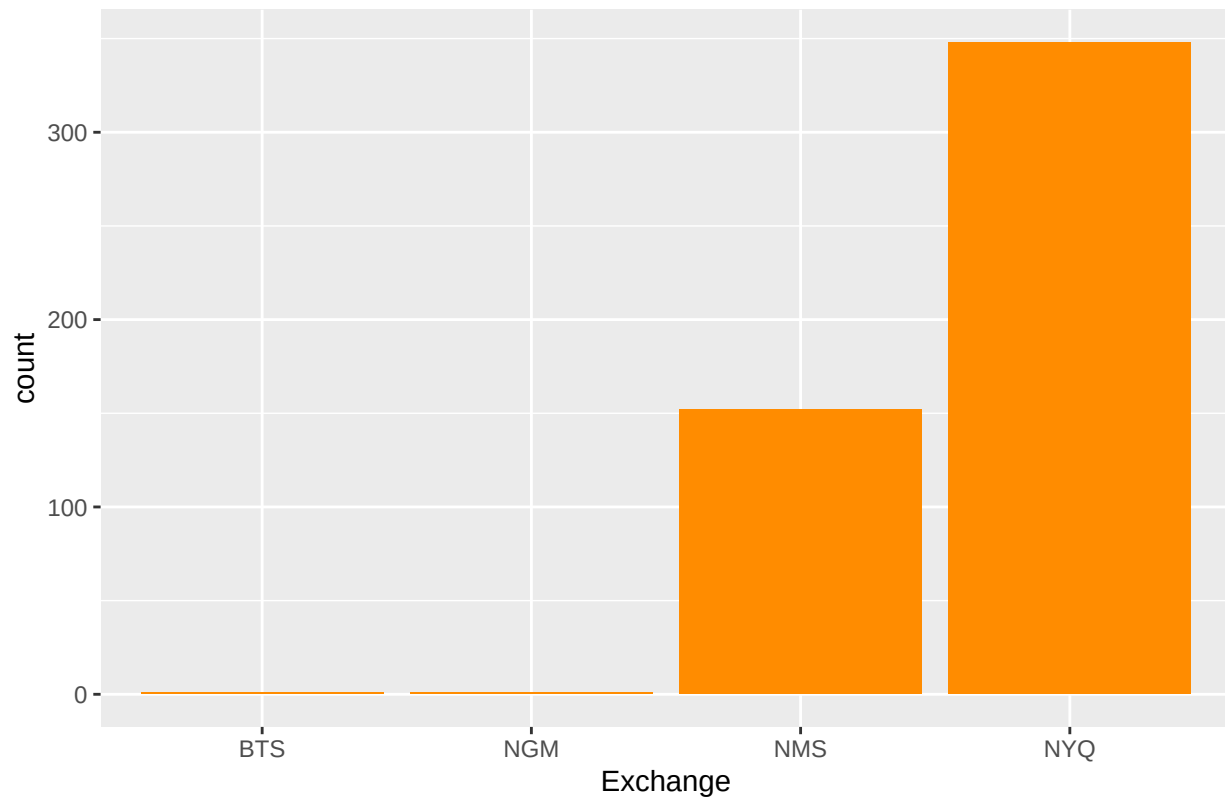
| skim_variable     | n_missing | complete_rate | mean         | sd           | p0           | p25          | p50          | p75          | p100         | hist |
|-------------------|-----------|---------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|------|
| Currentprice      | 0         | 1.00          | 2.178900e+02 | 4.892100e+02 | 9.4          | 6.855000e+01 | 1.83700e+02  | 2.276700e+02 | 8.27678e+03  |      |
| Marketcap         | 0         | 1.00          | 1.107222e+14 | 4.440074e+16 | 4.664099e+13 | 8.193341e+13 | 3.706396e+14 | 7.830565e+14 | 3.81682e+12  |      |
| Ebitda            | 29        | 0.94          | 7.045286e+10 | 1.024213e+10 | -            | 1.625000e+09 | 5.2748e+09   | 2.00000e+10  | 9.547e+11    |      |
| Revenuegrowth     | 3         | 0.99          | 7.000000e-02 | 1.800000e-01 | -0.6         | 0.000000e+00 | 5.000000e-01 | 1.100000e-01 | 1.63000e+00  |      |
| Fulltimeemployees | 9         | 0.98          | 5.782761e+05 | 1.395988e+05 | 28.0         | 1.020000e+05 | 2.169000e+05 | 5.510000e+05 | 2.140000e+06 |      |
| Weight            | 0         | 1.00          | 0.000000e+00 | 1.000000e-02 | 0.0          | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 7.000000e-02 |      |

```
ggplot(df, aes(x = fct_infreq(Sector))) +
  geom_bar(fill = "steelblue") +
  coord_flip() +
  labs(title = "Number of Companies per Sector", x = "Sector", y = "Count")
```



```
ggplot(df, aes(x = Exchange)) +  
  geom_bar(fill = "darkorange") +  
  labs(title = "Distribution of Stock Exchanges")
```

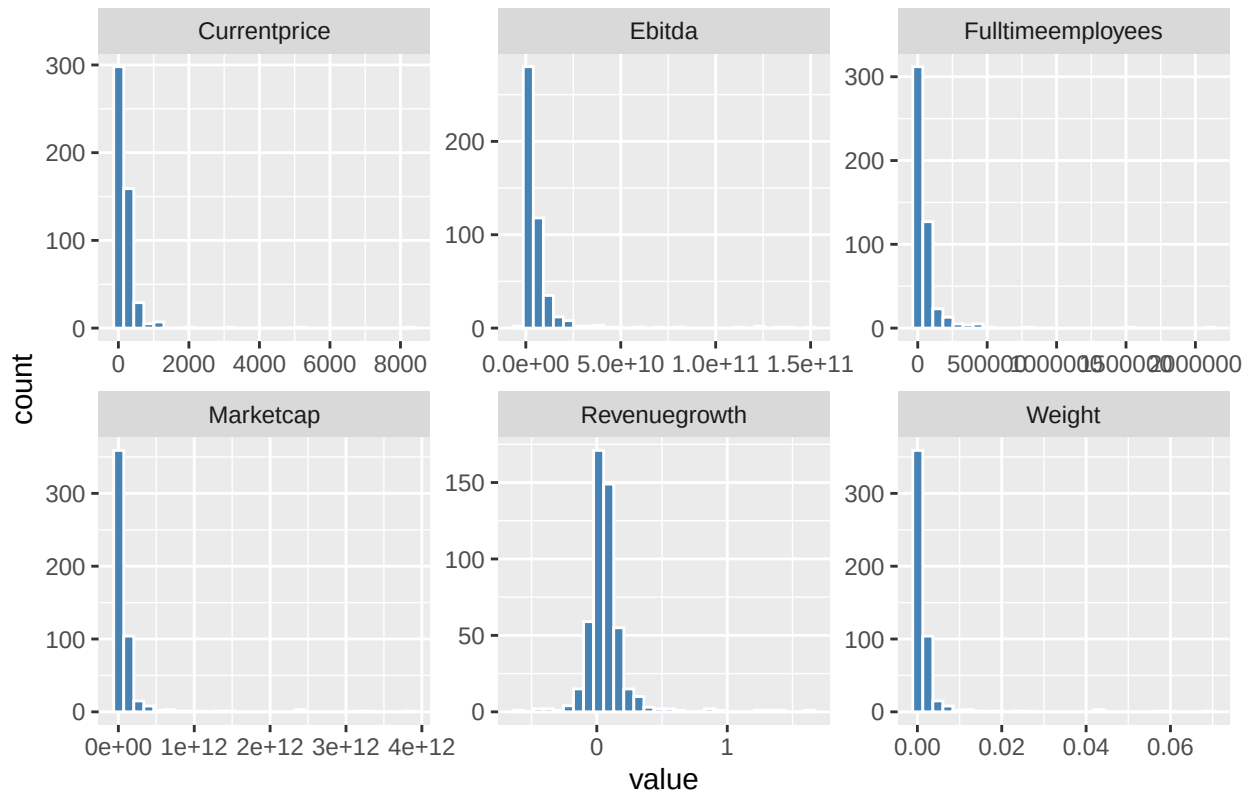
Distribution of Stock Exchanges



```
df %>%
  select(Currentprice, Marketcap, Ebitda, Revenuegrowth, Fulltimeemployees, Weight) %>%
  gather(key = "variable", value = "value") %>%
  ggplot(aes(x = value)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  facet_wrap(~variable, scales = "free") +
  labs(title = "Distributions of Numerical Variables")
```

```
## Warning: Removed 41 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

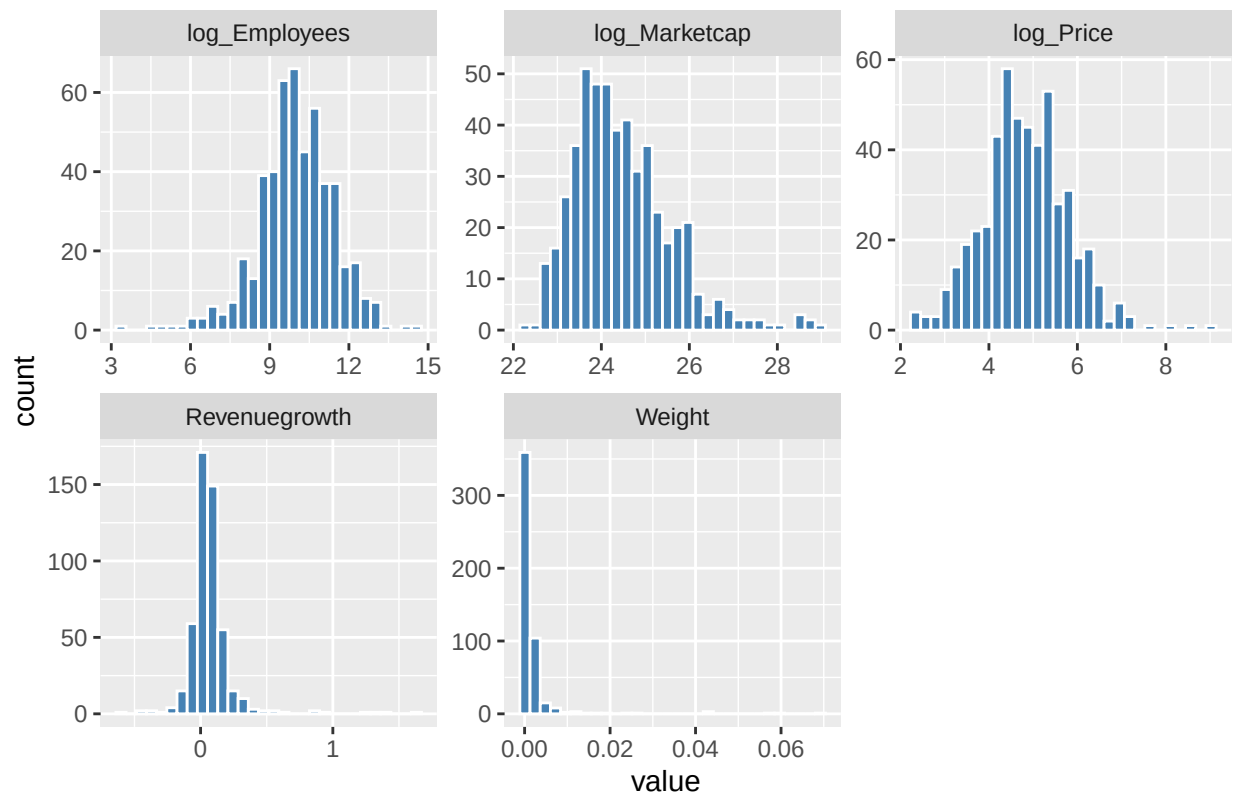
## Distributions of Numerical Variables



```
df %>%
  mutate(
    log_Marketcap = log1p(Marketcap),
    log_Employees = log1p(Fulltimeemployees),
    log_Price     = log1p(Currentprice)
  ) %>%
  select(log_Marketcap, log_Employees, log_Price, Revenuegrowth, Weight) %>%
  gather(key = "variable", value = "value") %>%
  ggplot(aes(x = value)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  facet_wrap(~variable, scales = "free") +
  labs(title = "Log Transformed Distributions")
```

```
## Warning: Removed 12 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

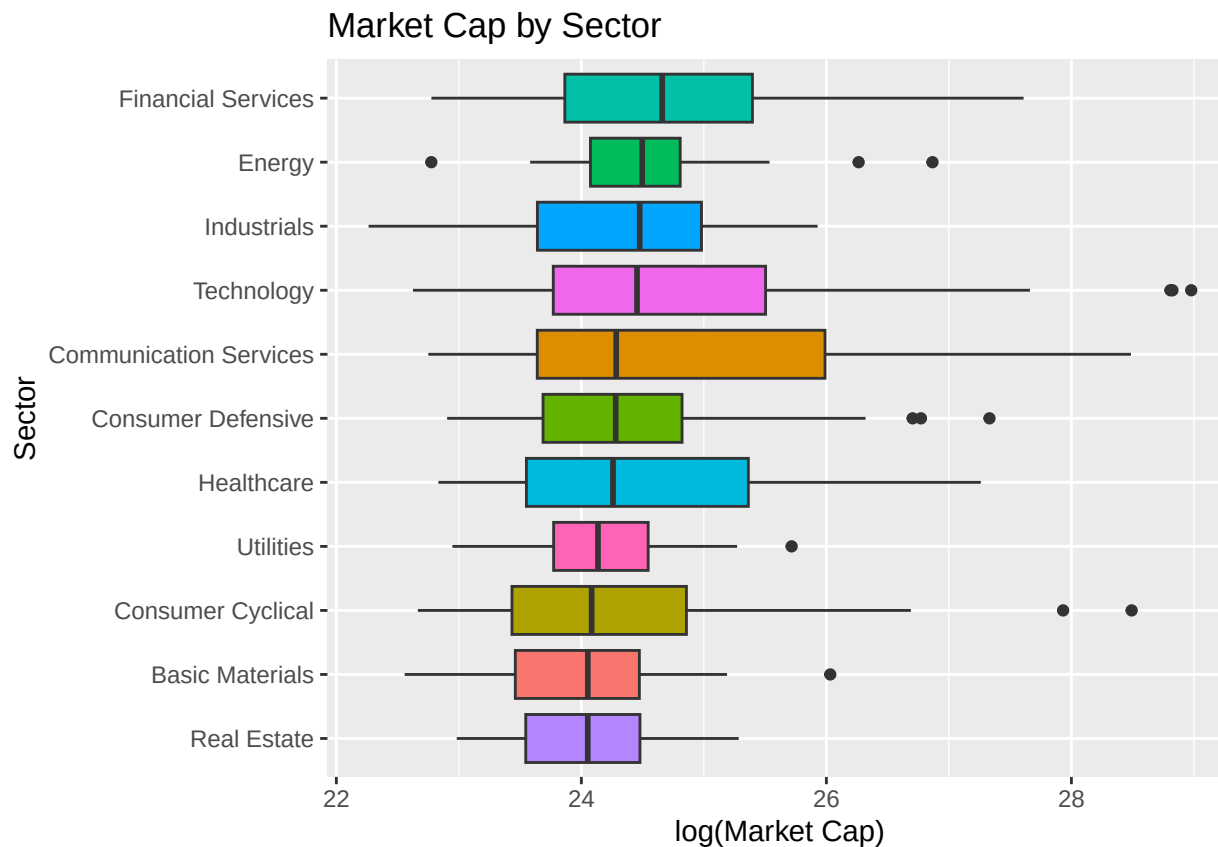
## Log Transformed Distributions



*# Relationship*

*# Marketcap by Sector*

```
ggplot(df, aes(x = fct_reorder(Sector, Marketcap, median), y = log1p(Marketcap), fill = Sector)) +  
  geom_boxplot(show.legend = FALSE) +  
  coord_flip() +  
  labs(title = "Market Cap by Sector", x = "Sector", y = "log(Market Cap)")
```

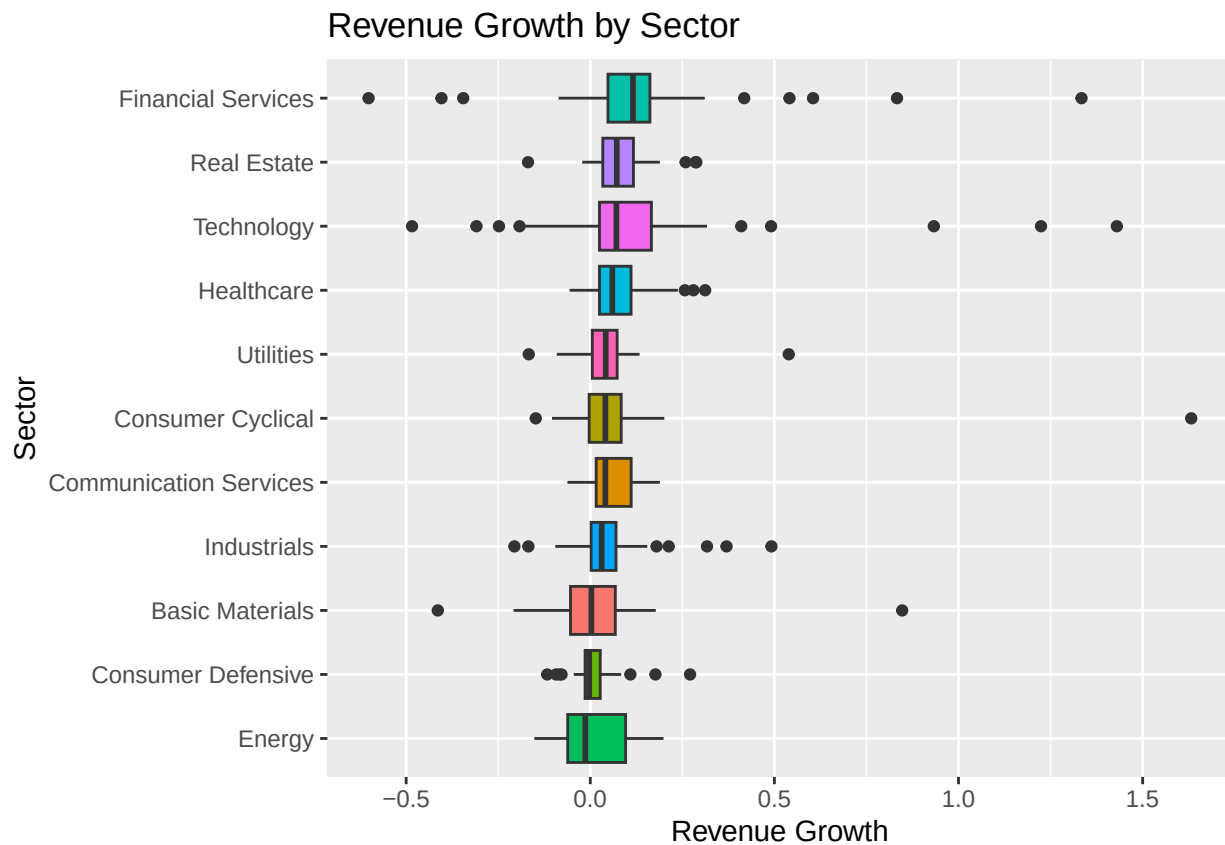


```
# Revenue Growth by Sector
ggplot(df, aes(x = fct_reorder(Sector, Revenuegrowth, median, na.rm = TRUE),
                             y = Revenuegrowth, fill = Sector)) +
  geom_boxplot(show.legend = FALSE) +
  coord_flip() +
  labs(title = "Revenue Growth by Sector", x = "Sector", y = "Revenue Growth")
```

```
## Warning: `fct_reorder()` removing 3 missing values.
## i Use `na.rm = TRUE` to silence this message.
## i Use `na.rm = FALSE` to preserve NAs.
```

```
## Warning: Removed 3 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

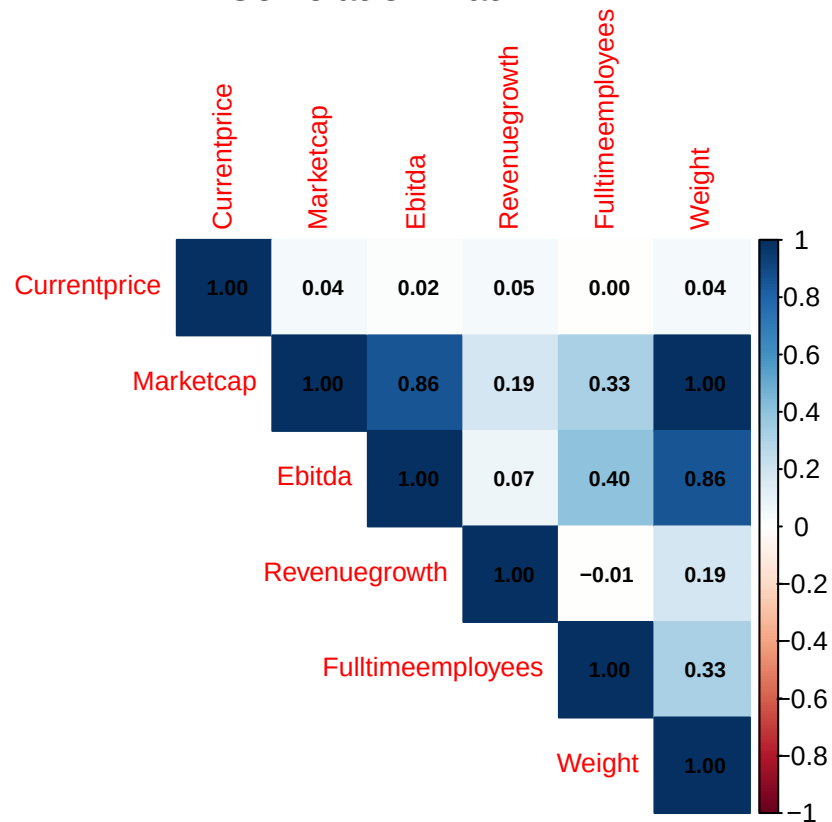




```
# Correlation matrix
num_df <- df %>%
  select(Currentprice, Marketcap, Ebitda, Revenuegrowth, Fulltimeemployees, Weight) %>%
  drop_na()

cor_matrix <- cor(num_df)
corrplot(cor_matrix, method = "color", type = "upper",
  tl.cex = 0.8, addCoef.col = "black", number.cex = 0.7,
  title = "Correlation Matrix", mar = c(0,0,1,0))
```

## Correlation Matrix



```
# Part c
# cleaning
library(tidyverse)
colSums(is.na(df))
```

```
##      Exchange      Symbol      Shortname      Longname
##           0           0           0           0
##      Sector      Industry      Currentprice      Marketcap
##           0           0           0           0
##      Ebitda      Revenuegrowth      City      State
##          29           3           0           0
##      Country Fulltimeemployees      Weight
##           0           9           0
```

```
df$Revenuegrowth[is.na(df$Revenuegrowth)] <- median(df$Revenuegrowth, na.rm = TRUE)

df$Fulltimeemployees[is.na(df$Fulltimeemployees)] <- median(df$Fulltimeemployees, na.rm = TRUE)

df <- df %>%
  group_by(Sector) %>%
  mutate(Ebitda = ifelse(is.na(Ebitda), median(Ebitda, na.rm = TRUE), Ebitda)) %>%
  ungroup()

df$State[is.na(df$State) | df$State == ""] <- "N/A"

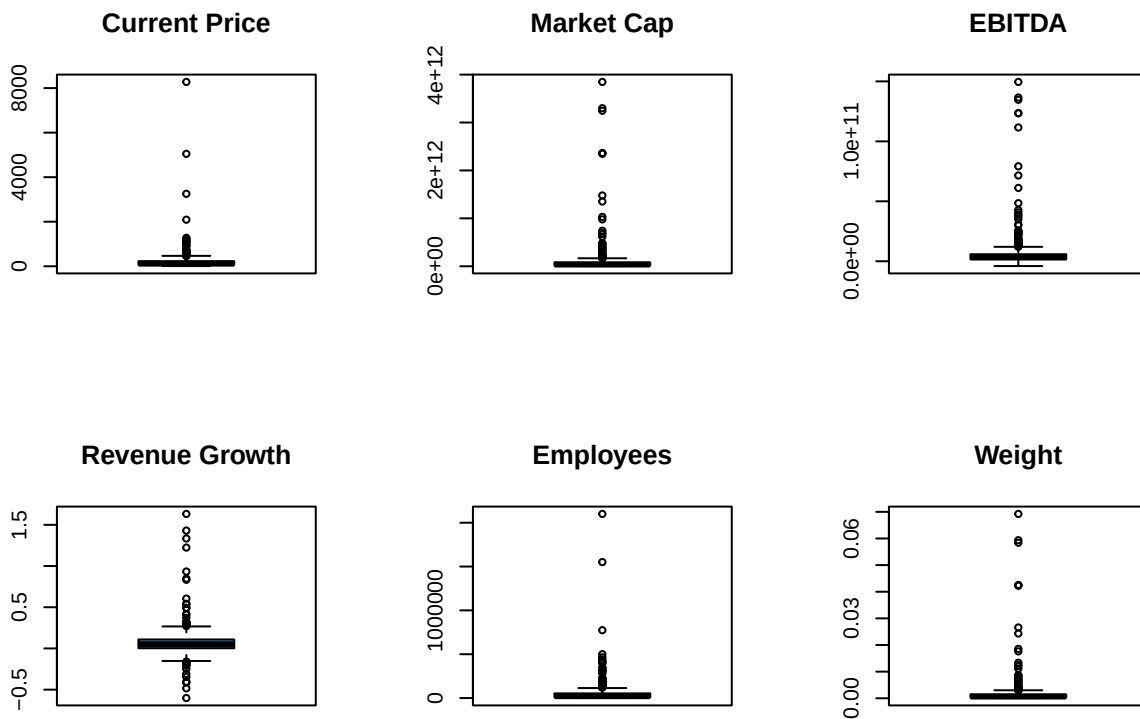
cat("Missing values after imputation:\n")
```

```
## Missing values after imputation:
```

```
colSums(is.na(df))
```

```
##      Exchange      Symbol      Shortname      Longname
##           0           0           0           0
##      Sector      Industry      Currentprice      Marketcap
##           0           0           0           0
##      Ebitda      Revenuegrowth      City      State
##           0           0           0           0
##      Country Fulltimeemployees      Weight
##           0           0           0
```

```
par(mfrow = c(2, 3))
boxplot(df$Currentprice, main = "Current Price", col = "steelblue")
boxplot(df$Marketcap, main = "Market Cap", col = "steelblue")
boxplot(df$Ebitda, main = "EBITDA", col = "steelblue")
boxplot(df$Revenuegrowth, main = "Revenue Growth", col = "steelblue")
boxplot(df$Fulltimeemployees, main = "Employees", col = "steelblue")
boxplot(df$Weight, main = "Weight", col = "steelblue")
```



```

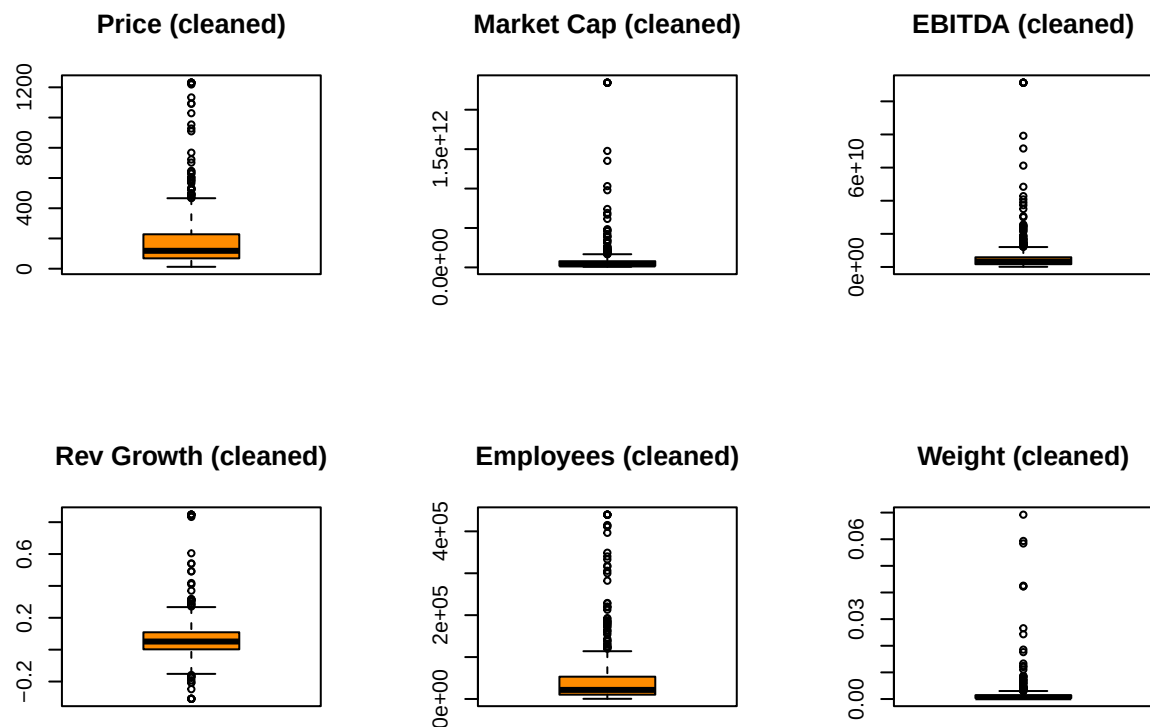
par(mfrow = c(1, 1))

winsorize <- function(x, lower = 0.01, upper = 0.99) {
  q <- quantile(x, probs = c(lower, upper), na.rm = TRUE)
  x <- ifelse(x < q[1], q[1], x)
  x <- ifelse(x > q[2], q[2], x)
  return(x)
}

df$Currentprice <- winsorize(df$Currentprice)
df$Marketcap <- winsorize(df$Marketcap)
df$Ebitda <- winsorize(df$Ebitda)
df$Revenuegrowth <- winsorize(df$Revenuegrowth)
df$Fulltimeemployees <- winsorize(df$Fulltimeemployees)

par(mfrow = c(2, 3))
boxplot(df$Currentprice, main = "Price (cleaned)", col = "darkorange")
boxplot(df$Marketcap, main = "Market Cap (cleaned)", col = "darkorange")
boxplot(df$Ebitda, main = "EBITDA (cleaned)", col = "darkorange")
boxplot(df$Revenuegrowth, main = "Rev Growth (cleaned)", col = "darkorange")
boxplot(df$Fulltimeemployees, main = "Employees (cleaned)", col = "darkorange")
boxplot(df$Weight, main = "Weight (cleaned)", col = "darkorange")

```



```

par(mfrow = c(1, 1))

```

```
df$Sector <- as.factor(df$Sector)
df$Industry <- as.factor(df$Industry)
df$Exchange <- as.factor(df$Exchange)
df$Country <- as.factor(df$Country)
df$State <- as.factor(df$State)

cat("\nFinal dataset dimensions:", dim(df), "\n")
```

```
##
## Final dataset dimensions: 502 15
```

```
skim(df)
```

Table 4: Data summary

|                        |      |
|------------------------|------|
| Name                   | df   |
| Number of rows         | 502  |
| Number of columns      | 15   |
| Column type frequency: |      |
| character              | 4    |
| factor                 | 5    |
| numeric                | 6    |
| Group variables        | None |

#### Variable type: character

| skim_variable | n_missing | complete_rate | min | max | empty | n_unique | whitespace |
|---------------|-----------|---------------|-----|-----|-------|----------|------------|
| Symbol        | 0         | 1             | 1   | 5   | 0     | 502      | 0          |
| Shortname     | 0         | 1             | 7   | 31  | 0     | 499      | 0          |
| Longname      | 0         | 1             | 6   | 47  | 0     | 499      | 0          |
| City          | 0         | 1             | 4   | 18  | 0     | 235      | 0          |

#### Variable type: factor

| skim_variable | n_missing | complete_rate | ordered | n_unique | top_counts                         |
|---------------|-----------|---------------|---------|----------|------------------------------------|
| Exchange      | 0         | 1             | FALSE   | 4        | NYQ: 348, NMS: 152, BTS: 1, NGM: 1 |
| Sector        | 0         | 1             | FALSE   | 11       | Tec: 82, Ind: 70, Fin: 67, Hea: 62 |
| Industry      | 0         | 1             | FALSE   | 114      | Uti: 23, Spe: 16, Sof: 15, Sem: 14 |
| State         | 0         | 1             | FALSE   | 42       | CA: 67, NY: 49, TX: 48, IL: 32     |
| Country       | 0         | 1             | FALSE   | 8        | Uni: 481, Ire: 10, Swi: 3, Uni: 3  |

#### Variable type: numeric

| skim_variable     | missing | complete_rate | mean         | sd           | p0           | p25          | p50          | p75          | p100          | hist |
|-------------------|---------|---------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------|------|
| Currentprice      | 0       | 1             | 1.904400e+02 | 2.020200e+02 | 2.52         | 6.855000e+01 | 1.183700e+02 | 2.276700e+02 | 1.0230620e+03 |      |
| Marketcap         | 0       | 1             | 1.039553e+28 | 2.828499e+28 | 1.225853e+28 | 1.323417e+28 | 3.706396e+28 | 7.830565e+28 | 2.3112881e+12 |      |
| Ebitda            | 0       | 1             | 6.675618e+10 | 1.000795e+10 | 527000.00    | 675625e+03   | 270000e+03   | 832783e+03   | 1.0912593e+11 |      |
| Revenuegrowth     | 0       | 1             | 7.000000e-02 | 1.500000e-01 | -0.31        | 0.000000e+00 | 0.000000e+00 | 1.100000e-01 | 8.500000e-01  |      |
| Fulltimeemployees | 0       | 1             | 5.082477e+07 | 7.76683e+04  | 18.68        | 1.035000e+02 | 169000e+03   | 318750e+03   | 4397500e+05   |      |
| Weight            | 0       | 1             | 0.000000e+00 | 1.000000e-02 | 0.00         | 0.000000e+00 | 0.000000e+00 | 0.000000e+00 | 7.000000e-02  |      |

```
#part d

# Data Preprocessing

library(tidyverse)

df_model <- df %>%
  select(-Symbol, -Shortname, -Longname, -City)

df_model <- df_model %>%
  mutate(
    Exchange_NYQ = as.integer(Exchange == "NYQ"),
    Exchange_NGM = as.integer(Exchange == "NGM"),
    Exchange_PCX = as.integer(Exchange == "PCX"),

    Country_NonUS = as.integer(Country != "United States"),

    State_grouped = fct_lump_n(State, n = 5)
  ) %>%

  mutate(dummy = 1) %>%
  pivot_wider(names_from = State_grouped, values_from = dummy,
    names_prefix = "State_", values_fill = 0) %>%
  select(-Exchange, -Country, -State)

normalize <- function(x) {
  return((x - min(x, na.rm = TRUE)) / (max(x, na.rm = TRUE) - min(x, na.rm = TRUE)))
}

df_model <- df_model %>%
  mutate(
    Currentprice_norm = normalize(Currentprice),
    Marketcap_norm = normalize(Marketcap),
    Ebitda_norm = normalize(Ebitda),
    Revenuegrowth_norm = normalize(Revenuegrowth),
    Fulltimeemployees_norm = normalize(Fulltimeemployees),
    Weight_norm = normalize(Weight)
  ) %>%

  select(-Currentprice, -Marketcap, -Ebitda,
```

```

    -Revenuegrowth, -Fulltimeemployees, -Weight)

cat("Preprocessed dataset dimensions:", dim(df_model), "\n")

## Preprocessed dataset dimensions: 502 19

str(df_model)

## tibble [502 x 19] (S3: tbl_df/tbl/data.frame)
##  $ Sector          : Factor w/ 11 levels "Basic Materials",...: 10 10 10 3 2 2 2 3 10 6 ...
##  $ Industry         : Factor w/ 114 levels "Advertising Agencies",...: 26 94 96 59 58 58 58 9 94
##  $ Exchange_NYQ     : int [1:502] 0 0 0 0 0 0 0 0 0 1 ...
##  $ Exchange_NGM     : int [1:502] 0 0 0 0 0 0 0 0 0 0 ...
##  $ Exchange_PCX     : int [1:502] 0 0 0 0 0 0 0 0 0 0 ...
##  $ Country_NonUS    : int [1:502] 0 0 0 0 0 0 0 0 0 0 ...
##  $ State_CA         : num [1:502] 1 1 0 0 1 1 1 0 1 0 ...
##  $ State_Other      : num [1:502] 0 0 1 1 0 0 0 0 0 1 ...
##  $ State_TX         : num [1:502] 0 0 0 0 0 0 0 1 0 0 ...
##  $ State_NY         : num [1:502] 0 0 0 0 0 0 0 0 0 0 ...
##  $ State_OH         : num [1:502] 0 0 0 0 0 0 0 0 0 0 ...
##  $ State_IL         : num [1:502] 0 0 0 0 0 0 0 0 0 0 ...
##  $ State_MA         : num [1:502] 0 0 0 0 0 0 0 0 0 0 ...
##  $ Currentprice_norm : num [1:502] 0.199 0.1 0.348 0.174 0.147 ...
##  $ Marketcap_norm   : num [1:502] 1 1 1 1 1 ...
##  $ Ebitda_norm      : num [1:502] 1 0.55 1 1 1 ...
##  $ Revenuegrowth_norm : num [1:502] 0.32 1 0.405 0.362 0.398 ...
##  $ Fulltimeemployees_norm : num [1:502] 0.3723 0.0664 0.518 1 0.4116 ...
##  $ Weight_norm      : num [1:502] 1 0.857 0.844 0.614 0.611 ...

head(df_model)

## # A tibble: 6 x 19
##   Sector Industry Exchange_NYQ Exchange_NGM Exchange_PCX Country_NonUS State_CA
##   <fct>  <fct>         <int>         <int>         <int>         <int>     <dbl>
## 1 Techno~ Consume~         0           0           0           0         1
## 2 Techno~ Semicon~         0           0           0           0         1
## 3 Techno~ Softwar~         0           0           0           0         0
## 4 Consum~ Interne~         0           0           0           0         0
## 5 Commun~ Interne~         0           0           0           0         1
## 6 Commun~ Interne~         0           0           0           0         1
## # i 12 more variables: State_Other <dbl>, State_TX <dbl>, State_NY <dbl>,
## #   State_OH <dbl>, State_IL <dbl>, State_MA <dbl>, Currentprice_norm <dbl>,
## #   Marketcap_norm <dbl>, Ebitda_norm <dbl>, Revenuegrowth_norm <dbl>,
## #   Fulltimeemployees_norm <dbl>, Weight_norm <dbl>

df_cluster <- df_model %>%
  select(-Sector, -Industry) %>%
  select(where(is.numeric))

cat("\nClustering dataset dimensions:", dim(df_cluster), "\n")

```

```
##
## Clustering dataset dimensions: 502 17
```

```
str(df_cluster)
```

```
## tibble [502 x 17] (S3: tbl_df/tbl/data.frame)
##  $ Exchange_NYQ      : int [1:502] 0 0 0 0 0 0 0 0 0 1 ...
##  $ Exchange_NGM      : int [1:502] 0 0 0 0 0 0 0 0 0 0 ...
##  $ Exchange_PCX      : int [1:502] 0 0 0 0 0 0 0 0 0 0 ...
##  $ Country_NonUS     : int [1:502] 0 0 0 0 0 0 0 0 0 0 ...
##  $ State_CA          : num [1:502] 1 1 0 0 1 1 1 0 1 0 ...
##  $ State_Other       : num [1:502] 0 0 1 1 0 0 0 0 0 1 ...
##  $ State_TX          : num [1:502] 0 0 0 0 0 0 0 0 1 0 ...
##  $ State_NY          : num [1:502] 0 0 0 0 0 0 0 0 0 0 ...
##  $ State_OH          : num [1:502] 0 0 0 0 0 0 0 0 0 0 ...
##  $ State_IL          : num [1:502] 0 0 0 0 0 0 0 0 0 0 ...
##  $ State_MA          : num [1:502] 0 0 0 0 0 0 0 0 0 0 ...
##  $ Currentprice_norm : num [1:502] 0.199 0.1 0.348 0.174 0.147 ...
##  $ Marketcap_norm    : num [1:502] 1 1 1 1 1 ...
##  $ Ebitda_norm       : num [1:502] 1 0.55 1 1 1 ...
##  $ Revenuegrowth_norm : num [1:502] 0.32 1 0.405 0.362 0.398 ...
##  $ Fulltimeemployees_norm: num [1:502] 0.3723 0.0664 0.518 1 0.4116 ...
##  $ Weight_norm       : num [1:502] 1 0.857 0.844 0.614 0.611 ...
```

```
# part e: clustering
```

```
library(tidyverse)
library(ggplot2)
library(factoextra)
```

```
## Welcome to factoextra!
## Want to learn more? See two factoextra-related books at https://www.datanovia.com/en/product/practice
```

```
# elbow method
```

```
set.seed(42)
```

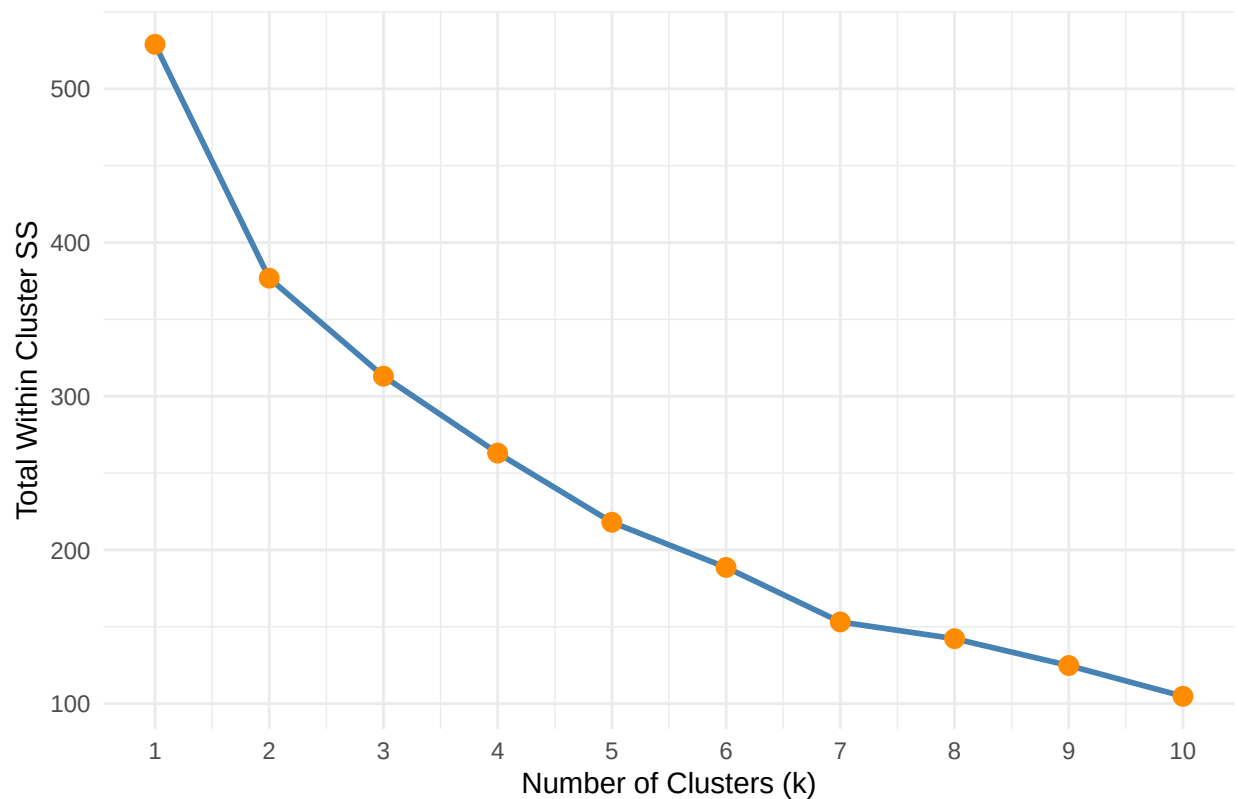
```
wss <- map_dbl(1:10, function(k) {
  kmeans(df_cluster, centers = k, nstart = 25)$tot.withinss
})
```

```
elbow_df <- data.frame(k = 1:10, wss = wss)
```

```
ggplot(elbow_df, aes(x = k, y = wss)) +
  geom_line(color = "steelblue", linewidth = 1) +
  geom_point(size = 3, color = "darkorange") +
  scale_x_continuous(breaks = 1:10) +
  labs(title = "Elbow Method: Choosing Number of Clusters",
       x = "Number of Clusters (k)", y = "Total Within Cluster SS") +
  theme_minimal()
```



## Elbow Method: Choosing Number of Clusters



```
# Fit K-Means with chosen k
```

```
k <- 8
```

```
set.seed(42)
```

```
kmeans_result <- kmeans(df_cluster, centers = k, nstart = 25)
```

```
df_model$Cluster <- as.factor(kmeans_result$cluster)
```

```
cat("Cluster sizes:\n")
```

```
## Cluster sizes:
```

```
table(df_model$Cluster)
```

```
##
```

```
##  1  2  3  4  5  6  7  8
```

```
## 22 49 180 21 32 61 70 67
```

```
pca_result <- prcomp(df_cluster, scale. = FALSE) # already normalized
```

```
pca_var <- summary(pca_result)$importance[2, 1:5]
```

```
cat("\nVariance explained by first 5 PCs:\n")
```

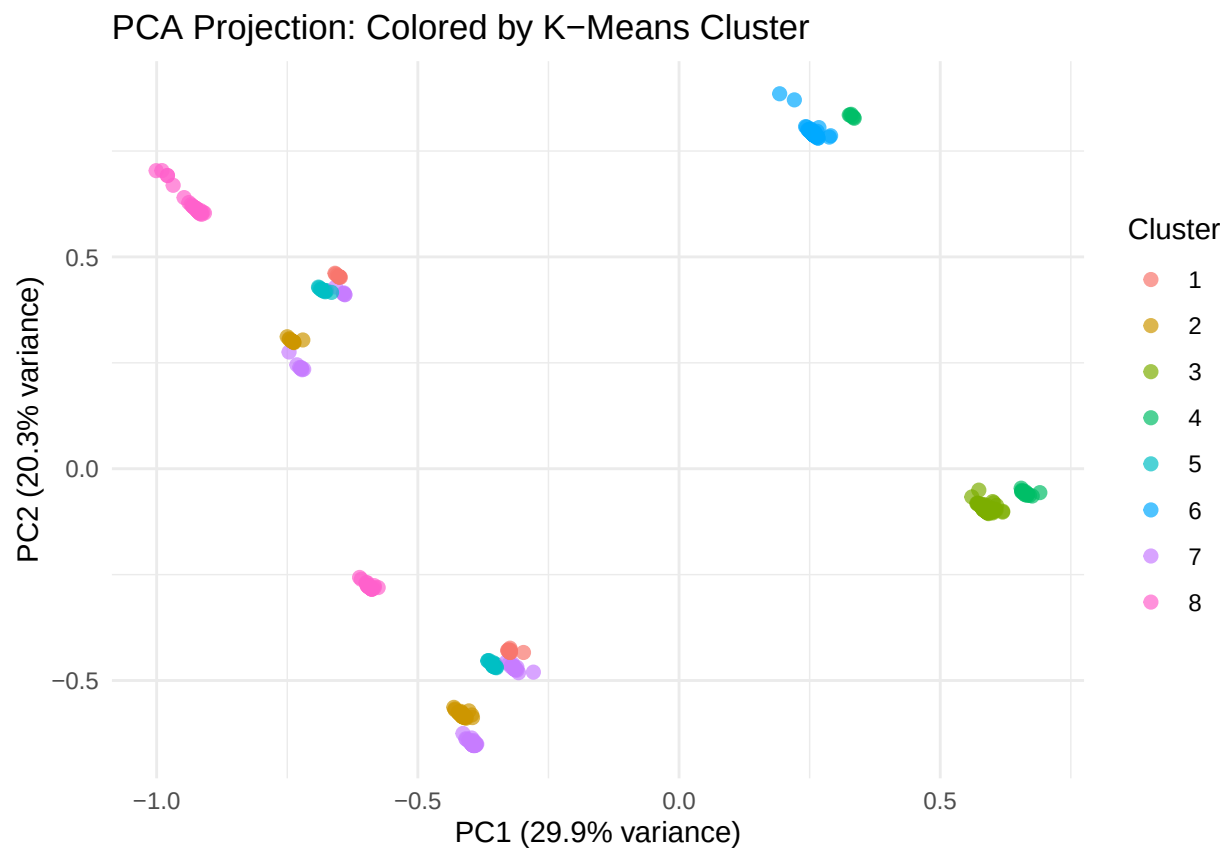
```
##
## Variance explained by first 5 PCs:
```

```
print(round(pca_var * 100, 2))
```

```
##   PC1   PC2   PC3   PC4   PC5
## 29.87 20.26 10.50  9.12  6.82
```

```
pca_df <- data.frame(
  PC1      = pca_result$x[, 1],
  PC2      = pca_result$x[, 2],
  Cluster  = df_model$Cluster,
  Sector   = df_model$Sector
)

ggplot(pca_df, aes(x = PC1, y = PC2, color = Cluster)) +
  geom_point(alpha = 0.7, size = 2) +
  labs(title = "PCA Projection: Colored by K-Means Cluster",
       x = paste0("PC1 (", round(pca_var[1] * 100, 1), "% variance)"),
       y = paste0("PC2 (", round(pca_var[2] * 100, 1), "% variance)")) +
  theme_minimal()
```

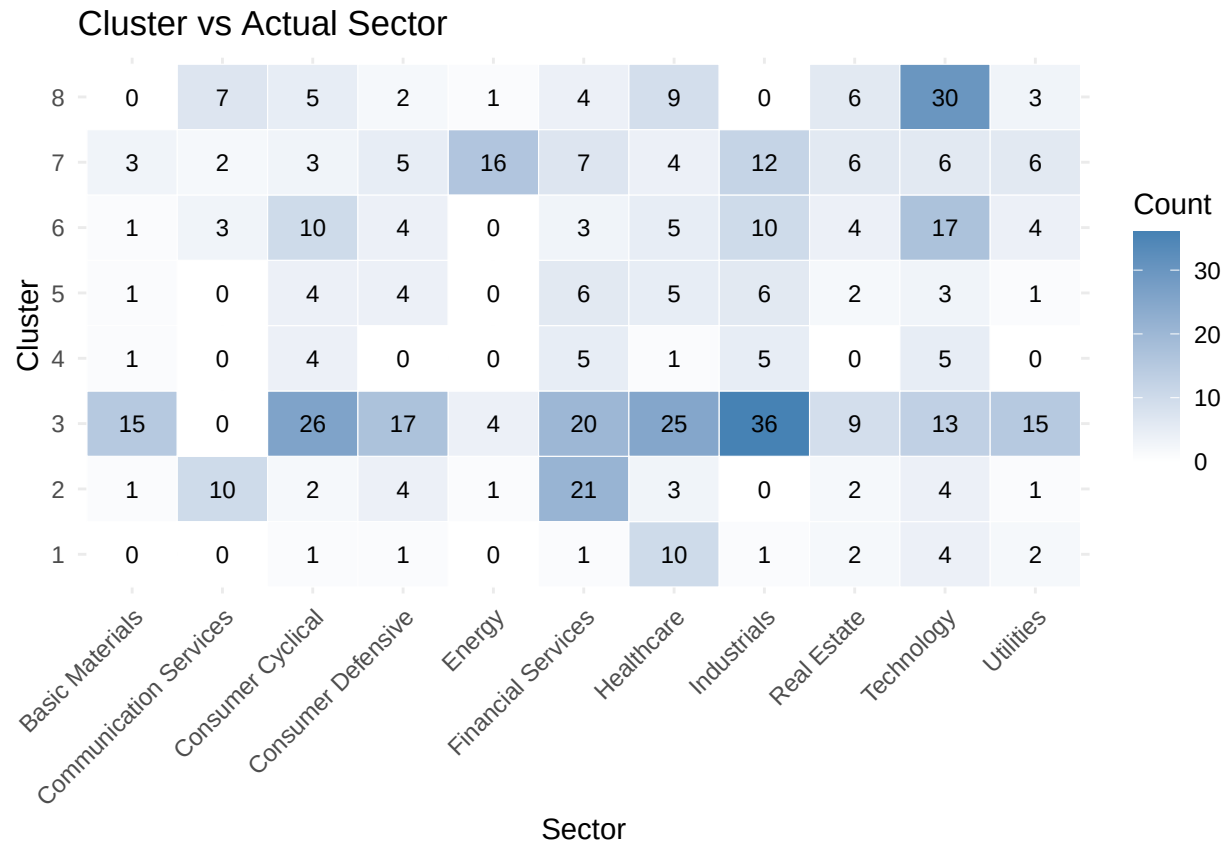


```
cluster_sector_table <- table(Cluster = df_model$Cluster,
                             Sector   = df_model$Sector)
print(cluster_sector_table)
```

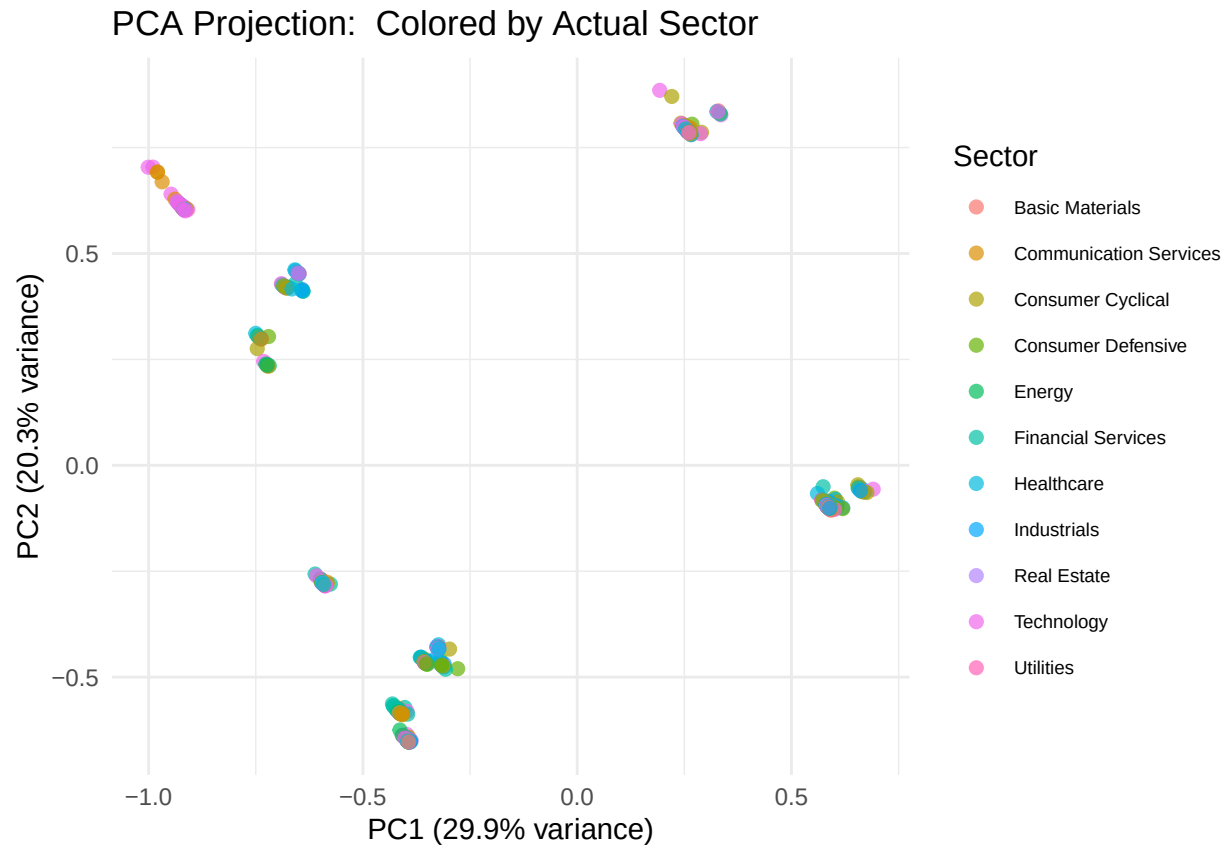
```
##      Sector
## Cluster Basic Materials Communication Services Consumer Cyclical
##      1          0          0          1
##      2          1         10          2
##      3         15          0         26
##      4          1          0          4
##      5          1          0          4
##      6          1          3         10
##      7          3          2          3
##      8          0          7          5
##      Sector
## Cluster Consumer Defensive Energy Financial Services Healthcare Industrials
##      1          1          0          1         10          1
##      2          4          1          21          3          0
##      3         17          4         20         25         36
##      4          0          0          5          1          5
##      5          4          0          6          5          6
##      6          4          0          3          5         10
##      7          5         16          7          4         12
##      8          2          1          4          9          0
##      Sector
## Cluster Real Estate Technology Utilities
##      1          2          4          2
##      2          2          4          1
##      3          9         13         15
##      4          0          5          0
##      5          2          3          1
##      6          4         17          4
##      7          6          6          6
##      8          6         30          3
```

```
cluster_sector_df <- as.data.frame(cluster_sector_table)

ggplot(cluster_sector_df, aes(x = Sector, y = Cluster, fill = Freq)) +
  geom_tile(color = "white") +
  geom_text(aes(label = Freq), size = 3) +
  scale_fill_gradient(low = "white", high = "steelblue") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Cluster vs Actual Sector",
       fill = "Count")
```



```
ggplot(pca_df, aes(x = PC1, y = PC2, color = Sector)) +
  geom_point(alpha = 0.7, size = 2) +
  labs(title = "PCA Projection: Colored by Actual Sector",
        x = paste0("PC1 (", round(pca_var[1] * 100, 1), "% variance)"),
        y = paste0("PC2 (", round(pca_var[2] * 100, 1), "% variance)")) +
  theme_minimal() +
  theme(legend.text = element_text(size = 7))
```



*# Part f Classification*

```
library(caret)
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##   lift
```

```
library(randomForest)
```

```
## randomForest 4.7-1.2
## Type rfNews() to see new features/changes/bug fixes.
##
## Attaching package: 'randomForest'
##
## The following object is masked from 'package:dplyr':
##
##   combine
##
## The following object is masked from 'package:ggplot2':
##
```

```
##      margin
```

```
library(rpart)
library(rpart.plot)

# Prepare data for classification

df_class <- df_model %>%
  select(-Cluster, -Industry) %>%
  mutate(Sector = as.factor(Sector))

table(df_class$Sector)
```

```
##
##      Basic Materials Communication Services      Consumer Cyclical
##              22              22              55
##      Consumer Defensive      Energy      Financial Services
##              37              22              67
##              Healthcare      Industrials      Real Estate
##              62              70              31
##              Technology      Utilities
##              82              32
```

```
# Train/Test split (80/20)
set.seed(42)
train_index <- createDataPartition(df_class$Sector, p = 0.8, list = FALSE)
train_data  <- df_class[train_index, ]
test_data   <- df_class[-train_index, ]

cat("Training rows:", nrow(train_data), "\n")
```

```
## Training rows: 405
```

```
cat("Testing rows: ", nrow(test_data), "\n")
```

```
## Testing rows: 97
```

```
# Model 1: Decision Tree
set.seed(42)

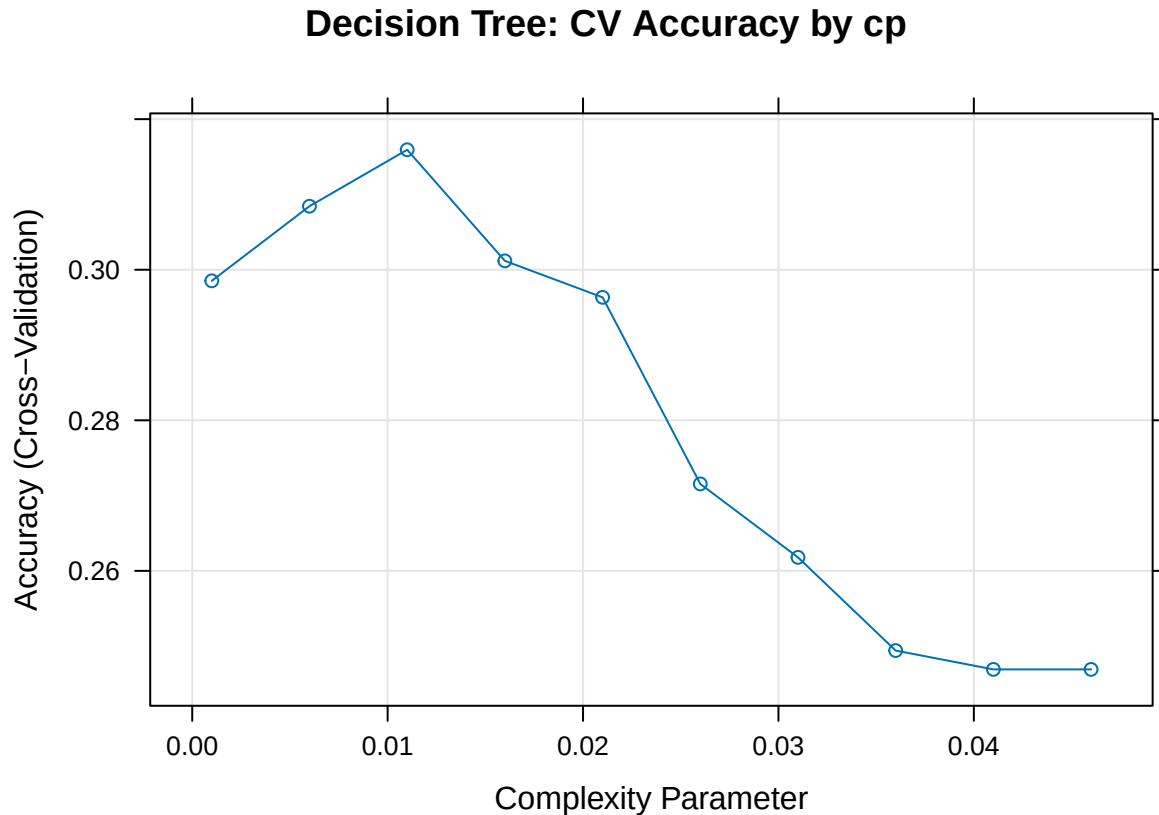
cp_grid <- expand.grid(cp = seq(0.001, 0.05, by = 0.005))

dt_model <- train(
  Sector ~ .,
  data      = train_data,
  method    = "rpart",
  trControl = trainControl(method = "cv", number = 5),
  tuneGrid  = cp_grid
)

cat("\nBest Decision Tree cp:", dt_model$bestTune$cp, "\n")
```

```
##  
## Best Decision Tree cp: 0.011
```

```
plot(dt_model, main = "Decision Tree: CV Accuracy by cp")
```

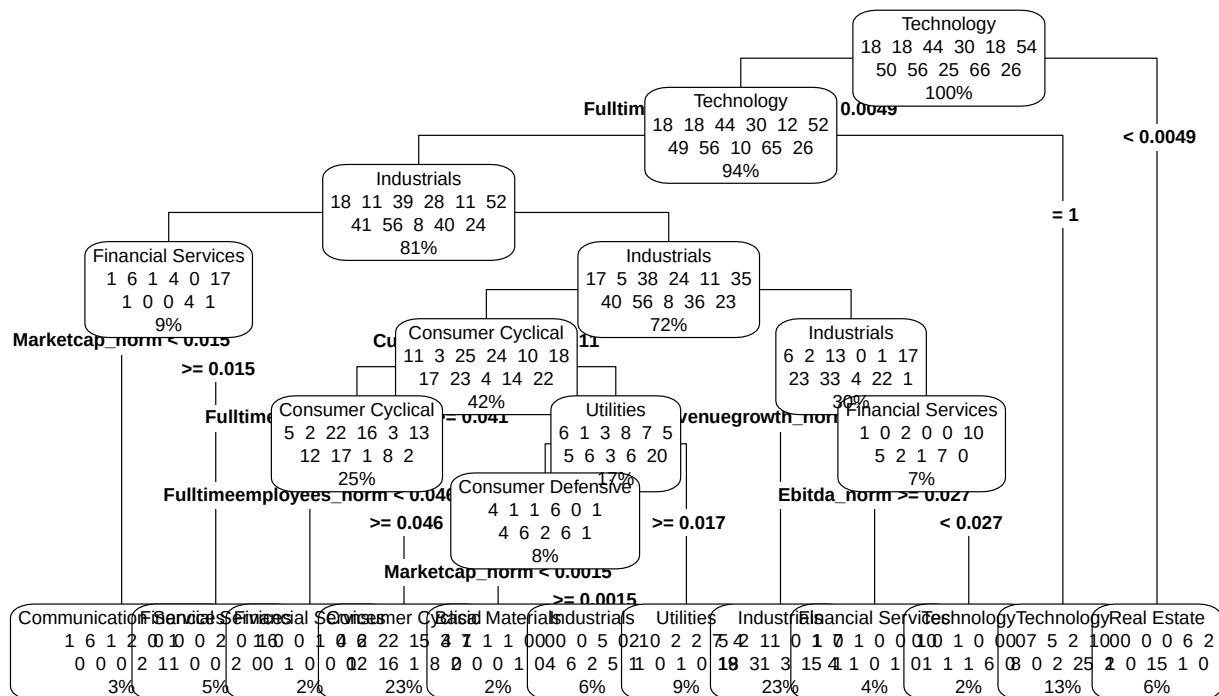


```
rpart.plot(dt_model$finalModel, type = 4, extra = 101,  
           main = "Decision Tree", cex = 0.6)
```

```
## Warning: All boxes will be white (the box.palette argument will be ignored) because  
## the number of classes in the response 11 is greater than length(box.palette) 6.  
## To silence this warning use box.palette=0 or trace=-1.
```

```
## Warning: labs do not fit even at cex 0.15, there may be some overplotting
```

## Decision Tree



```
dt_preds <- predict(dt_model, test_data)
dt_cm <- confusionMatrix(dt_preds, test_data$Sector)
cat("\nDecision Tree Accuracy:", round(dt_cm$overall["Accuracy"], 4), "\n")
```

```
##
## Decision Tree Accuracy: 0.3299
```

```
# Model 2: Random Forest
```

```
set.seed(42)
```

```
rf_grid <- expand.grid(mtry = c(2, 4, 6, 8, 10))
```

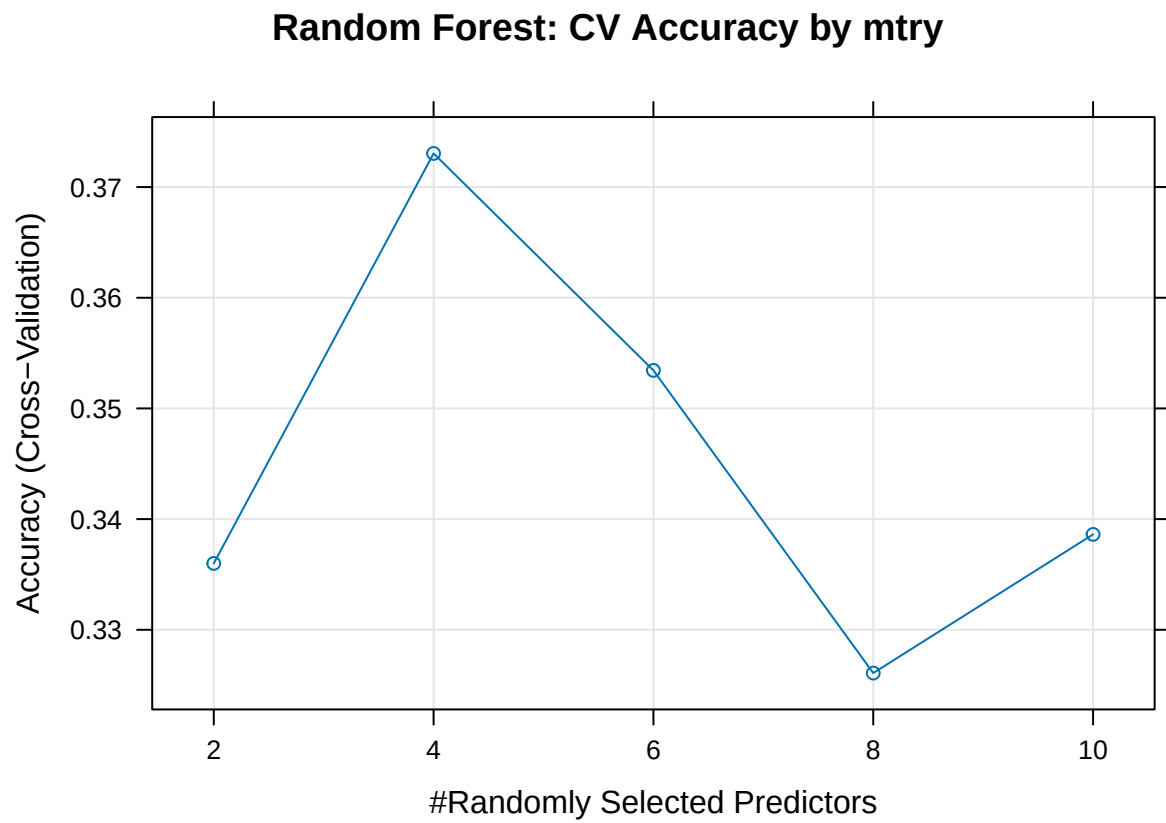
```
rf_model <- train(
  Sector ~ .,
  data      = train_data,
  method    = "rf",
  trControl = trainControl(method = "cv", number = 5),
  tuneGrid  = rf_grid,
  ntree     = 200
)
```

```
cat("\nBest Random Forest mtry:", rf_model$bestTune$mtry, "\n")
```

```
##
## Best Random Forest mtry: 4
```

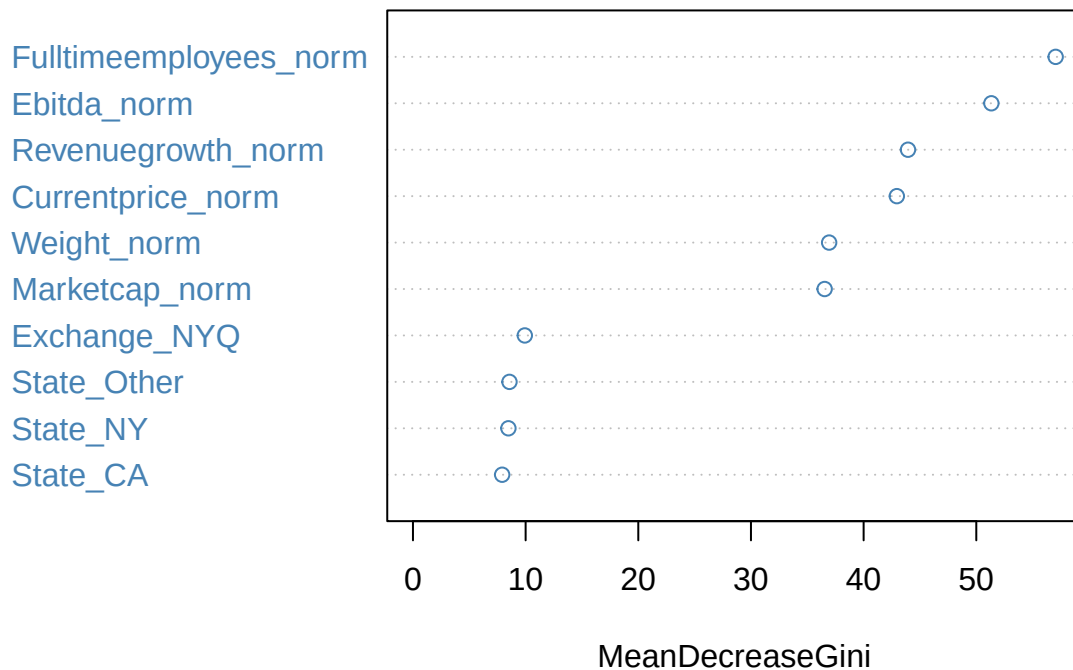


```
plot(rf_model, main = "Random Forest: CV Accuracy by mtry")
```



```
varImpPlot(rf_model$finalModel, main = "Random Forest: Variable Importance",  
           n.var = 10, col = "steelblue")
```

## Random Forest: Variable Importance



```
rf_preds <- predict(rf_model, test_data)
rf_cm <- confusionMatrix(rf_preds, test_data$Sector)
cat("\nRandom Forest Accuracy:", round(rf_cm$overall["Accuracy"], 4), "\n")
```

```
##
## Random Forest Accuracy: 0.3814
```

```
# Compare both models
cat("\n Model Comparison \n")
```

```
##
## Model Comparison
```

```
cat("Decision Tree Accuracy: ", round(dt_cm$overall["Accuracy"], 4), "\n")
```

```
## Decision Tree Accuracy: 0.3299
```

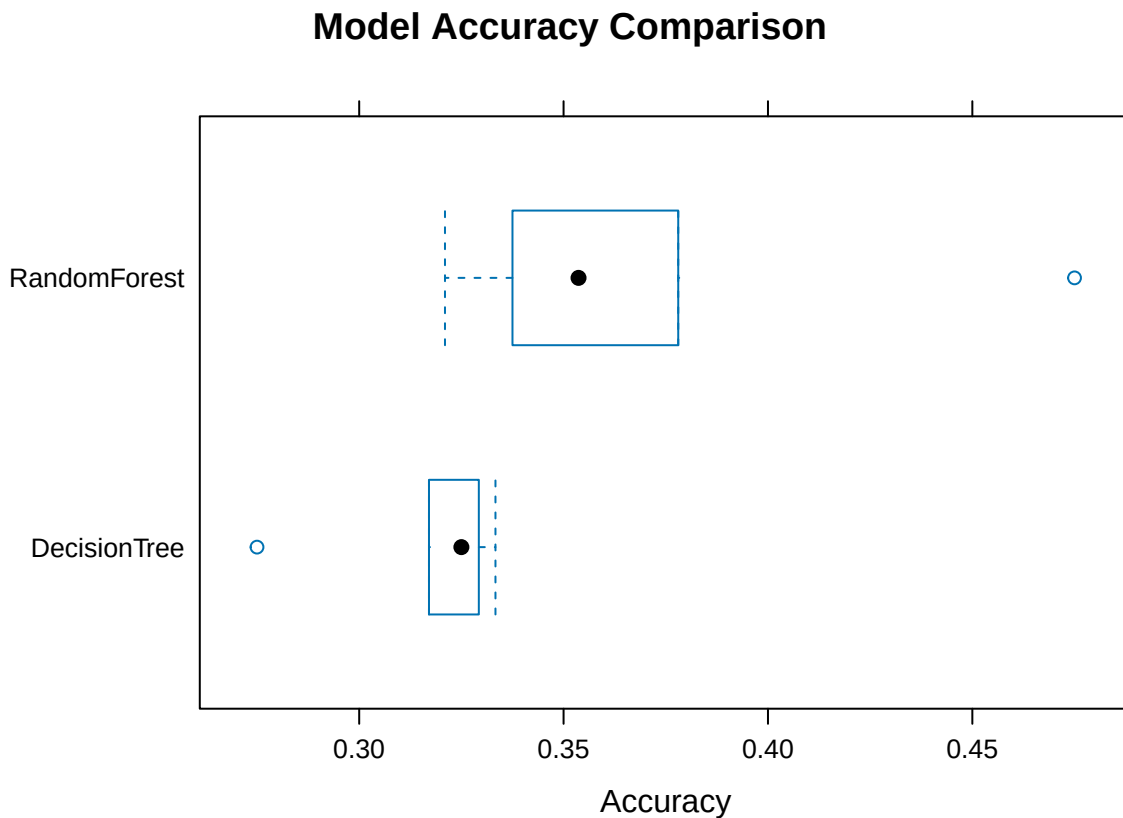
```
cat("Random Forest Accuracy: ", round(rf_cm$overall["Accuracy"], 4), "\n")
```

```
## Random Forest Accuracy: 0.3814
```

```
resamps <- resamples(list(DecisionTree = dt_model, RandomForest = rf_model))
summary(resamps)
```

```
##
## Call:
## summary.resamples(object = resamps)
##
## Models: DecisionTree, RandomForest
## Number of resamples: 5
##
## Accuracy
##           Min.    1st Qu.    Median     Mean   3rd Qu.     Max. NA's
## DecisionTree 0.2750000 0.3170732 0.3250000 0.315935 0.3292683 0.3333333    0
## RandomForest 0.3209877 0.3375000 0.3536585 0.373039 0.3780488 0.4750000    0
##
## Kappa
##           Min.    1st Qu.    Median     Mean   3rd Qu.     Max. NA's
## DecisionTree 0.1754043 0.2214310 0.2377894 0.2255194 0.2459417 0.2470305    0
## RandomForest 0.2308356 0.2480936 0.2567128 0.2863960 0.2925055 0.4038325    0
```

```
bwplot(resamps, metric = "Accuracy", main = "Model Accuracy Comparison")
```



*# Part G: Evaluation*

```

library(caret)
library(pROC)

## Type 'citation("pROC")' for a citation.
##
## Attaching package: 'pROC'
##
## The following objects are masked from 'package:stats':
##
##     cov, smooth, var

# Bin sectors into 2 groups

growth_sectors <- c("Technology", "Consumer Cyclical", "Communication Services",
                    "Financial Services", "Industrials")

df_class2 <- df_class %>%
  mutate(Sector_binary = as.factor(
    ifelse(Sector %in% growth_sectors, "Growth", "Defensive")
  ))

table(df_class2$Sector_binary)

##
## Defensive      Growth
##      206      296

# Train/Test split
set.seed(42)
train_index2 <- createDataPartition(df_class2$Sector_binary, p = 0.8, list = FALSE)
train2 <- df_class2[train_index2, ] %>% select(-Sector)
test2 <- df_class2[-train_index2, ] %>% select(-Sector)

# Rebuild RF on binary
set.seed(42)
rf_binary <- train(
  Sector_binary ~ .,
  data      = train2,
  method    = "rf",
  trControl = trainControl(method = "cv", number = 5,
                           classProbs = TRUE,
                           summaryFunction = twoClassSummary),
  tuneGrid  = expand.grid(mtry = 4),
  metric    = "ROC",
  ntree     = 200
)

# Predict on test set
rf_preds2 <- predict(rf_binary, test2)
rf_probs2 <- predict(rf_binary, test2, type = "prob")[, "Growth"]

# Confusion Matrix

```

```
cm2 <- confusionMatrix(rf_preds2, test2$Sector_binary, positive = "Growth")
print(cm2)
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  Defensive Growth
## Defensive      21      15
## Growth         20      44
##
##           Accuracy : 0.65
##           95% CI : (0.5482, 0.7427)
##       No Information Rate : 0.59
##       P-Value [Acc > NIR] : 0.1313
##
##           Kappa : 0.2628
##
##  Mcnemar's Test P-Value : 0.4990
##
##           Sensitivity : 0.7458
##           Specificity : 0.5122
##       Pos Pred Value : 0.6875
##       Neg Pred Value : 0.5833
##           Prevalence : 0.5900
##       Detection Rate : 0.4400
##       Detection Prevalence : 0.6400
##       Balanced Accuracy : 0.6290
##
##       'Positive' Class : Growth
##
```

```
cat("\n Confusion Matrix \n")
```

```
##
## Confusion Matrix
```

```
print(cm2$table)
```

```
##           Reference
## Prediction  Defensive Growth
## Defensive      21      15
## Growth         20      44
```

```
TP <- cm2$table["Growth", "Growth"]
FP <- cm2$table["Growth", "Defensive"]
FN <- cm2$table["Defensive", "Growth"]
TN <- cm2$table["Defensive", "Defensive"]
```

```
precision <- TP / (TP + FP)
recall    <- TP / (TP + FN)
f1        <- 2 * (precision * recall) / (precision + recall)
```

```

accuracy <- (TP + TN) / sum(cm2$table)

cat("TP:", TP, " FP:", FP, " FN:", FN, " TN:", TN, "\n")

## TP: 44  FP: 20  FN: 15  TN: 21

cat("Precision:", round(precision, 4), "\n")

## Precision: 0.6875

cat("Recall:   ", round(recall, 4), "\n")

## Recall:    0.7458

cat("F1 Score: ", round(f1, 4), "\n")

## F1 Score:  0.7154

cat("Accuracy: ", round(accuracy, 4), "\n")

## Accuracy:  0.65

# ROC curve
roc_obj <- roc(test2$Sector_binary, rf_probs2, levels = c("Defensive", "Growth"))

## Setting direction: controls < cases

plot(roc_obj,
     col      = "steelblue",
     lwd      = 2,
     main     = paste0("ROC Curve: Random Forest (AUC = ",
                       round(auc(roc_obj), 3), ")"),
     xlab     = "False Positive Rate (1: Specificity)",
     ylab     = "True Positive Rate (Sensitivity)",
     abline(a = 0, b = 1, lty = 2, col = "gray")

```

